



THE BOXES

Research dossier

A drama set inside NASA Mission Control across the four days of Apollo 13, from the oxygen-tank rupture to splashdown in April 1970.

The research reconstructs the four-day crisis of Apollo 13 entirely inside NASA Manned Spacecraft Center Building 30 in April 1970. Ground teams work through continuous shift rotations to manage life-support failures, power depletion, and manual orbital calculations. The film centers on the flight controllers whose prior testing compromises and procedural oversights led directly to the oxygen tank rupture. The story follows their internal guilt as they sit at the consoles attempting to recover the crew they endangered. The drama stays confined to the physical infrastructure of the operations floor, backroom support trenches, and surrounding Clear Lake facilities.

Depth	Scout	
Research confidence	77%	
Coverage	85%	
Status	—	
Boxes	13	
Evidence fragments	241	
Distinct sources	184	
Distinct domains	72	
Contradictions	0 conflict, 0 context	
Research passes	4	
Generated	2026-09-09 13:37 UTC	
Project id	demo	

THE BOXES

What the research found, in plain language, box by box. Every claim below is traceable to the sources listed under it.

PRESS CORPS INFRASTRUCTURE

99% 8 items · 5 domains

Building 1 housed the central auditorium press center adjacent to the mission operations wing. Television networks set up fixed camera stands, monitor arrays, and teletype machinery to file stories. Banks of public payphones lined the facility corridors for reporters covering the crisis. Press representatives worked in assigned auditorium seats facing formal agency briefing podiums.

- Building on a Mission: The Auditorium and External Relations Office Building - NASA · 2023-09-28
- News Media Handbook · 2005-02-01
- The display/control complex of the Manned Space Mission Control ...
- The Press Site at KSC · 2006-11-29
- Mission Control Center (MCC) - Apollo 13 - Fourth (4th ...
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS

98% 12 items · 6 domains

Manned Spacecraft Center security functioned under federal property guidelines alongside the Houston Police Department. Federal records track campus access controls, guard posts, and site security transmittals during 1970. Area law enforcement maintained routine monitoring of civilian areas and regional political protests during the Vietnam War era. Security details guarded exterior access points into critical flight infrastructure buildings.

- Records - Houston Police Department
- [PDF] Police Practices in Houston, Texas - U.S. Commission on Civil Rights
- Records transmittal and receipt from NASA Manned Spacecraft Center to Federal Records Center, GSA, Fort Worth, Texas for Record Group 255
- Historical Documents - Office of the Historian · 2026-06-29
- Building on a Mission: The Houston Mission Control Center - NASA · 2023-09-23
- Research in Vietnam War Records
- Houston Police Department - Public Information Requests
- Vietnam War: The War at Home | National Archives · 2022-06-17
- News Release Archives
- Houston Police Department
- Transparency

REAL-TIME TELEMETRY DATA FLOW · emergent

97% 14 items · 7 domains

Spacecraft telemetry moved from deep-space tracking stations through the Manned Space Flight Network to Houston mainframes. Goddard Space Flight Center routed raw radio metric data into the Building 30 processing complex. Ground computers processed incoming bitstreams to drive console displays and wall projection screens. Tracking data updated flight dynamics plotters to confirm spacecraft trajectory in real time.

- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT · 2002-10-31
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ... · 2005-06-09
- Apollo Era Technical Description
- Deep Space Network - NASA Science · 2025-01-16
- Integrated telemetry-computer system
- Telemetry and Command (T&C) Systems · 2026-03-13
- https://tda.jpl.nasa.gov/progress_report/42-25/25C.PDF
- NATIONAL AERONAUTICS AND SPACE ADMINISTRATION 1971 15,
- 1970 December 27 ~(C
- [PDF] Real-Time Flight Test Data Distribution and Display
- Network Space Deep The 7, Progress'Report.42-40
- SECTION 7 CRT DISPLAY PAGES
- -

ENGINEERING DIVISION DEMOGRAPHICS

97%  15 items · 5 domains

The flight control division consisted of young civilian engineers, university graduates, and military veterans. Personnel rosters documented a workforce predominantly in their twenties and thirties operating under strict functional titles. Controllers held specialized positional assignments including guidance officers, flight dynamics officers, and environmental specialists. The group operated under flat managerial oversight during real-time flight emergencies.

- Information Summaries Astronaut Fact Book - NASA · 2013-04-08
- Astrona · 2008-05-23
- Apollo 13 - NASA · 2023-03-29
- Apollo 13 Archives - NASA
- List of NASA's flight control positions - Wikipedia · 2026-06-21
- Flight controller - Wikipedia · 2026-06-29
- of
- Documentary Spotlights Apollo-Era NASA Flight Controllers – Houston Public Media · 2017-04-15
- Msc Personnel
- Apollo 13 II - NASA · 2023-07-06
- biographical summaries

CONSOLE HARDWARE ARCHITECTURE

96%  16 items · 9 domains

Building 30 housed Mission Operations Control Rooms on its second and third floors. Controllers monitored operations through cathode-ray tube displays, illuminated push-button arrays, and central projection screens. Pneumatic tube systems linked the main floor consoles to backroom support stations for physical document delivery. Console panels contained specialized lighting schematics and numeric readouts mapped to specific spacecraft systems.

- Apollo Mission Control Center Restoration - NASA · 2026-04-08
- NASA's restored Apollo Mission Control is a slice of '60s life, frozen in amber - Ars Technica · 2025-05-16
- Apollo Mission Control Photos - Space Center Houston · 2022-09-23
- MOCR - Manned Spaceflight Operations Association · 2018-03-13
- Orbiter Displays Reference Guide (OI-34)
- AS-508 MISOO FIU A IN
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15
- COMMAND AND SERVICE MODULE CONTROLS AND ...
- mission operations center: Topics by Science.gov
- CRT Reference Summary OI-27
- Microsoft Word - Final Drft, MCC 2010-12-30
- [PDF] Apollo Mission Control Restoration - NASA
- GitHub - johnmknigh/MOCR-Console-Replica-Suite: Period ...
- The display/control complex of the Manned Space Mission Control ...

COMMUNICATIONS LOOP ACOUSTICS

96%  17 items · 11 domains

Ground communications relied on dedicated audio loops dividing the flight director, capsule communicator, trench specialists, and recovery teams. Voice recordings captured precise operational cadences mixed with line hiss and static. Controllers maintained flat, procedural tones while transcribing air-to-ground exchanges during telemetry dropouts. Specialized audio restoration projects preserved these multi-channel recordings across the full duration of the mission.

- Apollo 13, transcription du vol
- Mission Control Audio Channel
- Apollo 13 in Real Time
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · 2020-11-23
- Apollo 13 Transcripts
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · 2026-06-15
- Mission Transcripts: Mercury, Gemini, and Apollo - NASA · 2026-06-15
- NASA Audio Highlight Reels, Soundbites and Launch Sounds
- Apollo 13 Audio

- NASA, university researchers release 19000 hours of ... · 2018-07-25
- Apollo 11 : NASA : Free Download, Borrow, and Streaming · 2010-12-07
- Recovering Audio from the Apollo Moon Missions
- Apollo 11 - Technical Air-to-Ground Voice Transcription

CLEAR LAKE COMMUTING

95%  12 items · 7 domains

NASA personnel commuted into the isolated Clear Lake complex via regional Texas access highways. Parking lots fronted standard federal security checkpoints and gate access lanes. Controllers drove era-standard personal sedans and station wagons onto the government property. Local police departments and site security details regulated movement into the campus perimeter.

- 309 Clear lake Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 19 Clear lake area, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Clear Lake - Police - City of Houston
- 5 Lct area, 1970 s Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- ROAD SURFACES AROUND NASA, NASA Technology Images — PICRYL - Public Domain Media Search Engine
- 60 Years Ago: NASA Selects Houston as Site for New Manned Spacecraft Center - NASA · 2023-08-30
- Microsoft Word - If99-6723_final-AVT-152-005_NASA_Experience_w-Pogo_in_Human_Spaceflight_Vehicles_Final.doc
- 104 Clear lake, Texas Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Clear lake, Houston Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- City of Houston eGovernment Center > About Houston > NASA
- CLEAR LAKE No. 81
- MAP OF THE CLEAR LAKE AREA

TECHNICAL CRISIS VOCABULARY

94%  10 items · 6 domains

Flight controllers communicated using dense telemetry shorthand and system acronyms. Operational logs recorded operational terms including GUIDO, RETRO, FIDO, and LM malfunction procedures. Floor personnel used abbreviations to report pressure decays, cryogenic levels, and thruster duty cycles. Communication transcripts show ground teams shortening complex physical states into terse operational phrases.

- [PDF] - 3 GLOSSARY OF ACRONYMS AND ABBREVIATIONS - Ibiblio
- GLOSSARY OF TERMS · 2010-01-27
- Appendix B: Acronyms, Abbreviations, and Symbols - NASA · 2026-04-22
- Apollo 13 LM Malfunction Procedures
- [PDF] _.-_m MANNED SPACECRAFT CENTER - NASA
- NASA-Apollo-Mission-Control-Acronyms. ...
- Space Transportation System and Associated Payloads: Glossary, Acronyms, and Abbreviations
- Space Flight Technical Language Handbook
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

FACILITY LIGHTING PROTOCOLS

94%  10 items · 9 domains

The operations room maintained strict low-level ambient illumination to keep front projection screens legible. Fluorescent overhead banks were dimmed or turned off, leaving consoles lit by internal indicators and CRT display glow. Measured foot-candle levels across the floor remained low to preserve controller night vision and screen contrast. Desks relied on small localized task lamps to illuminate manual logbooks and clipboards.

- PPF to Foot-candle Reference Table
- Foot-candle - Wikipedia · 2026-03-08
- FOOTCANDLE LIGHT GUIDE - Lighting Design Lab · 2013-07-29
- Foot Candle Calculator - Lighting Requirements & Illuminance ...
- https://www.usmd.edu/usm/adminfinance/Facilities/forms/interior_lighting_fluorescent.pdf
- Lighting Illumination Calculator - everycalculators.com
- Apollo Flight Controller 101: Every console explained - Ars Technica · 2019-12-25

- 3.PDF
- 20F Technology Committee On-Screen Light Measurement ...

MANUAL CALCULATION TOOLS · emergent

89%  14 items · 5 domains

Controllers used manual computation tools when primary telemetry systems failed. Engineers calculated spacecraft maneuvers using five-inch Pickett slide rules, graphing paper, and grease pencils. Hand calculators and technical plotting boards served to verify emergency burn times and battery margins. Technical manuals provided paper-based malfunction procedures for guidance, navigation, and control systems.

- Apollo Command Module Block II Primary Guidance, Navigation, and Control System Manual · 2018-07-27
- Quasi Official Apollo 13 Paper September 8.ppt · 2009-09-08
- - · 2004-09-22
- Slide Rule, 5-inch, Pickett N600-ES, Apollo 13 | National Air and Space Museum
- Don't Know What a Slide Rule is For | National Air and Space Museum
- STUDY OF CONTROL COMPUTERS FOR ...
- AS-508 MISOO FIU A IN
- E-2448 Rev 1, Users' Guide to Apollo GN&CS Major Modes and Routines
- 1970 Slide Rule
- Slide rule vintage hi-res stock photography and images
- ORBITAL TESTING REQUIREMENTS FOR GUIDANCE AND CONTROL DEVICES

CONTROL ROOM CONSUMABLES

62%  38 items · 13 domains

Consoles and backrooms filled with operational consumables during around-the-clock rotations. Controllers consumed instant coffee from paper cups alongside vending machine snacks and takeout cartons. Deep glass ashtrays filled with discarded tobacco butts across desk surfaces. Official center food services provided basic provisions to sustain uninterrupted floor coverage.

- 93 Space food, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 120 Food containers, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Food Services - NASA · 2026-09-04
- NASA Facts
- [PDF] LYNDON B. JOHNSON SPACE CENTER
- Apollo Mission Control Photos - Space Center Houston
- 130 Food container, Crew Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 132 Food container, Activities Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 223 Spacecrew supplies Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo Mission Control Restoration - NASA
- APOLLO EXPERIENCE REPORT
- Apollo 13 Flight Journal - Mission Documents
- N76 12690 · 2004-01-23
- 163 Food container, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Space food, Food Images: PICRYL - Public Domain Media Search Engine Public Domain Search

SHIFT ROTATION ATTIRE

58%  27 items · 12 domains

Controllers worked eighty-four consecutive hours under escalating crisis conditions. Archival imagery shows white short-sleeve dress shirts wrinkling, sweating, and staining across multiple rotations. Neckties hung loose around unbuttoned collars as shifts blended into emergency monitoring sessions. Flight consoles collected discarded garments, badge lanyards, and ash residue across the four days.

- Mission Control, Houston, April 13, 1970 - NASA · 2015-04-13
- Apollo 13: The Successful Failure · 2020-04-06
- Apollo 13 - Wikipedia · 2026-08-13
- Apollo 13

- File:1970 Mission Control Apollo 13.jpg - Wikimedia Commons
- How Mission Control saved Apollo 13 - BBC Sky at Night Magazine
- PHOTOGRAPHS WRITTEN HISTORICAL AND ...
- 165 Flight controllers, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 992 Controllers Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 252 Consoles, Control Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo 13, NASA Apollo program. NASA public domain image... — PICRYL - Public Domain Media Search Engine
- Project Apollo Archive's albums | Flickr
- skandalon
- 43 Flight clothing, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- I found an archive of every picture taken by the astronauts ...

SUPPORT STAFF TRENCH DYNAMICS · emergent

26%  32 items · 13 domains

Staff Support Rooms operated directly behind the main control room to supply raw computational analysis. Support teams worked at dense clusters of desks tied directly to the frontline console positions. The real-time auxiliary computing facility processed complex orbital adjustments and spacecraft power limits. Engineers transferred handwritten calculations and printed readouts forward to the flight director.

- Apollo experience report - real-time auxiliary computing facility development · 2002-04-17
- Mission Control, Houston · 2006-05-01
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · 2012-11-14
- The Trench: The People at Mission Control | American Experience | Official Site | PBS · 2017-12-28
- Diagram of NASA Mission Control | National Air and Space Museum
- Apollo Mission Control Center Restoration · 2026-04-08
- The newly restored Apollo Mission Control Room : r/nasa
- Background Data (Process Overview)
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ...
- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15 · 2002-02-05
- Mission Operations Directorate New Employee Orientation
- Apollo 13
- Apollo 13 (film)
- Apollo 13, We Have a Solution: Part 2 - IEEE Spectrum · 2021-07-25

CONTRADICTIONS

Where two sources disagree. Similarity alone never decides this; each pair below was read and judged.

None found.

REFERENCE REEL

The strongest material, cut into a sequence a director can read.

00:00 Establishing Mission Control

Archival materials document the control room environment during Apollo 13 in Houston.

- Mission Control, Houston, April 13, 1970 - NASA · nasa.gov
- Apollo Mission Control Center Restoration - NASA · nasa.gov · 2026-04-08
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · arstechnica.com · 2012-11-14
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · skyatnightmagazine.com · 2026-06-15

00:30 Multi-Channel Communications

Restored multi-track audio channels record simultaneous conversations across flight controller consoles.

- Mission Control Audio Channel · apolloinrealtime.org
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · everydayastronaut.com · 2020-11-23
- Apollo 13 in Real Time · apolloinrealtime.org

01:00 Air-to-Ground Crisis Logs

Technical flight transcripts record critical communications following the oxygen tank failure.

- Apollo 13, transcription du vol · capcomespace.net
- Apollo 13 Transcripts · apollojournals.org
- Apollo 13 Audio · dn801309.us.archive.org

01:30 Extended Crisis Strain

Real-time records capture the extended four-day rescue operation through splashdown.

- Apollo 13 in Real Time · apolloinrealtime.org
- Apollo 13 - NASA · nasa.gov

DEPARTMENTS

The same boxes, grouped by which crew briefs off them, with what each one actually contains.

Script

COMMUNICATIONS LOOP ACOUSTICS · 17 items

Ground communications relied on dedicated audio loops dividing the flight director, capsule communicator, trench specialists, and recovery teams. Voice recordings captured precise operational cadences mixed with line hiss and static. Controllers maintained flat, procedural tones while transcribing air-to-ground exchanges during telemetry dropouts. Specialized audio restoration projects preserved these multi-channel recordings across the full duration of the mission.

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Casting

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Art direction

CONSOLE HARDWARE ARCHITECTURE · 16 items

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PRESS CORPS INFRASTRUCTURE · 8 items

Building 1 housed the central auditorium press center adjacent to the mission operations wing. Television networks set up fixed camera stands, monitor arrays, and teletype machinery to file stories. Banks of public payphones lined the facility corridors for reporters covering the crisis. Press representatives worked in assigned auditorium seats facing formal agency briefing podiums.

SUPPORT STAFF TRENCH DYNAMICS · 32 items

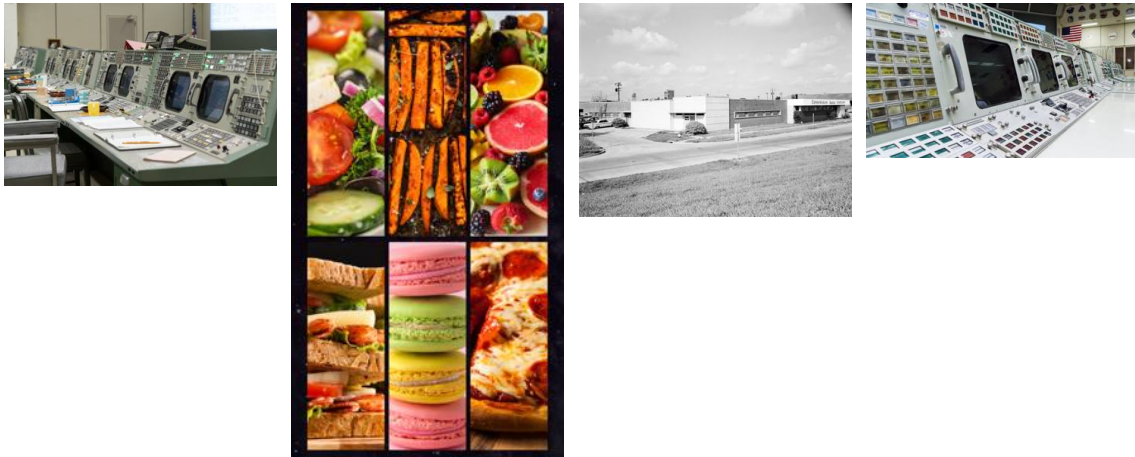
Staff Support Rooms operated directly behind the main control room to supply raw computational analysis. Support teams worked at dense clusters of desks tied directly to the frontline console positions. The real-time auxiliary computing facility processed complex orbital adjustments and spacecraft power limits. Engineers transferred handwritten calculations and printed readouts forward to the flight director.

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Cinematography

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Locations

CLEAR LAKE COMMUTING · 12 items

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PRIOR ART

65 existing films surveyed for a similar premise, seeded from: space crisis, mission control, engineers, survival, 1970s.

Apollo 13 (1995)

exploited mechanical flaw in the oxygen system · victim · triumphant · professional duty overcomes disaster · crew rescued after safe splashdown

similarity 0.72 · <https://www.themoviedb.org/movie/568>

Salyut-7 (2017)

exploited electrical flaw in station hardware · investigator · heroic · loyalty to comrades redeems political coercion · station restored to operational capability

similarity 0.66 · <https://www.themoviedb.org/movie/438740>

The Right Stuff (1983)

exploited psychological flaw in institutional hubris · bystander · satirical · individual audacity transcends bureaucratic manipulation · pioneering era closes with success

similarity 0.66 · <https://www.themoviedb.org/movie/9549>

The Real Right Stuff (2020)

exploited historical record shipped as archival media · bystander · documentary · collective effort secures a national achievement · historical missions conclude successfully

similarity 0.64 · <https://www.themoviedb.org/movie/753325>

First Man (2018)

exploited emotional flaw of unexpressed grief · victim · somber · private sorrow finds release through professional endurance · mission completed with silent personal reconciliation

similarity 0.64 · <https://www.themoviedb.org/movie/369972>

The Martian (2015)

exploited scientific ingenuity shipped as standard equipment · victim · optimistic · relentless problem-solving affirms human resilience · stranded astronaut successfully retrieved

similarity 0.63 · <https://www.themoviedb.org/movie/286217>

Hidden Figures (2016)

exploited systemic bias flaw in workplace hierarchy · investigator · inspiring · competence forces institutional integration · ground calculations validate the orbital trajectory

similarity 0.62 · <https://www.themoviedb.org/movie/381284>

Love (2011)

exploited psychological isolation flaw in prolonged confinement · victim · melancholic · alienation deepens into surreal acceptance · isolated subject achieves transcendence

similarity 0.58 · <https://www.themoviedb.org/movie/54320>

First Man into Space (1959)

exploited biological mutation flaw caused by cosmic radiation · perpetrator · tragic · reckless ambition brings monstrous ruin · dying monster provides crucial scientific data

similarity 0.56 · <https://www.themoviedb.org/movie/28955>

The Devil's Mouth (2026)

exploited physical vulnerability flaw in a trapped environment · victim · dread · survival pressure fractures ethical solidarity · diminished survivors escape the predator

similarity 0.55 · <https://www.themoviedb.org/movie/1481343>

Santo vs. the Martian Invasion (1967)

exploited alien invasion feature shipped as advanced weaponry · investigator · camp · physical heroism preserves terrestrial security · alien invaders suffer total defeat

similarity 0.55 · <https://www.themoviedb.org/movie/80766>

It All Adds Up - The Making of Hidden Figures (2017)

exploited promotional commentary shipped as retrospective access · bystander · celebratory · historical recovery honors overlooked labor · filmmakers validate historical relevance

similarity 0.55 · <https://www.themoviedb.org/movie/1533402>

Where this stands, and where it does not yet

Originality is claimed only relative to the titles above, never absolutely.

The narrative adopts the perpetrator point of view inside Mission Control, focusing on flight controllers whose previous procedural oversights caused the technical failure.

None of the listed titles present the disaster strictly through the culpability of the ground controllers.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Frame the central flight controller as the engineer who signed off on the compromised oxygen tank specifications.

The narrative operates as an institutional tragedy ending in the irreversible destruction of administrative careers despite the survival of the flight crew.

None of the listed films conclude with bureaucratic self-destruction alongside a successful rescue.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Show ground command securing splashdown while an internal audit terminates their program authority.

The narrative adopts a tone of relentless bureaucratic dread, focusing exclusively on ground support personnel paralyzed by statistical inevitability.

None of the listed historical or space flight films use an sustained tone of pure procedural dread without heroic relief.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Confine the perspective entirely to telemetry analysts convinced their models predict crew asphyxiation.

RESEARCH LOG

What each pass searched, found, and did next. Process detail, not required reading.

PASS 000

114 sources · 133 fragments · 40 via Extract

coverage 0% → 82% confidence 0% → 77%

CONSOLE HARDWARE ARCHITECTURE — Apollo Mission Operations Control Room MOCR console schematics Philco Ford CRT pneumatic tube; NASA MSC Building 30 MOCR console hardware display controls layout technical report

COMMUNICATIONS LOOP ACOUSTICS — Apollo 13 flight director loop capcom audio recordings transcript April 1970; NASA Apollo 13 internal communications voice loop trench backroom audio tapes archive

ENGINEERING DIVISION DEMOGRAPHICS — Apollo 13 flight controllers roster biographies educational background military service; NASA MSC Flight Operations Division personnel demographics flight controllers 1970

SHIFT ROTATION ATTIRE — NASA Mission Control Apollo 13 high resolution archival photographs controllers MOCR April 1970; Apollo 13 mission control room shift rotation candid photos controllers clothing rumpled

CONTROL ROOM CONSUMABLES — NASA Mission Control center Building 30 cafeteria vending coffee brands tobacco 1960s 1970; Apollo 13 mission operations control room ashtrays coffee cups takeout food archival photos

TECHNICAL CRISIS VOCABULARY — Apollo 13 telemetry acronyms abbreviations flight controller jargon mission operational terms; NASA Apollo flight controller terminology life support EPS EECOM telemetry shorthand handbook

FACILITY LIGHTING PROTOCOLS — MSC Building 30 Mission Operations Control Room lighting levels foot-candles display illumination specifications; Apollo MOCR rear projection screens ambient illumination lighting protocols technical manual

CLEAR LAKE COMMUTING — NASA Manned Spacecraft Center Clear Lake employee parking lot gate security vehicles 1970; Clear Lake Houston NASA Road 1 commute personnel parking cars April 1970 historical photos

PRESS CORPS INFRASTRUCTURE — MSC Building 1 auditorium news center Apollo 13 press corps floor plan teletype; Apollo 13 press room television camera positions NASA Manned Spacecraft Center newsroom setup 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS — NASA Manned Spacecraft Center security procedures Vietnam War antiwar protests 1970; MSC Houston perimeter security police civil defense coordination NASA complex April 1970 records

→ investigate CONTROL ROOM CONSUMABLES (score 0.31)

PASS 001

36 sources · 34 fragments · 12 via Extract

coverage 82% → 82% confidence 77% → 76%

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SHIFT ROTATION ATTIRE — Apollo 13 Mission Operations Control Room high resolution archival photos controllers April 1970; NASA flight controllers clothing shirts ties fatigue Apollo 13 photographs shift changes

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Control coffee cups cigarette brands food containers April 1970; NASA MOCR consoles food drink ashtrays vending snacks Apollo missions archive photos

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Mission Control Staff Support Room backroom layout procedures trench; Apollo 13 RETRO FIDO backroom trajectory calculation procedures support staff organizational chart

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 002

38 sources · 32 fragments · 12 via Extract

coverage 82% → 83% confidence 76% → 75%

opened: MANUAL CALCULATION TOOLS, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room MOCR coffee cups cigarettes food trash photographs April 1970; NASA MSC Building 30 Mission Control consumables vending machines ashtrays brands 1970

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Staff Support Room backroom layout procedures Trench FIDO GUIDO flight dynamics; Apollo 13 Mission Operations Control Room Staff Support Room organizational hierarchy communication loop

MANUAL CALCULATION TOOLS — Apollo 13 mission control slide rules calculators grease pencils trajectory manual calculation tools; NASA MOCR Apollo flight controller slide rule model Friden calculator graphing paper inventory 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 003

41 sources · 33 fragments · 12 via Extract

coverage 83% → 85% confidence 75% → 76%

opened: REAL-TIME TELEMETRY DATA FLOW, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room food coffee cigarettes photos; NASA Mission Control Center building 30 consumables vending coffee cups 1970

SUPPORT STAFF TRENCH DYNAMICS — Apollo 13 Staff Support Room backroom flight dynamics trench organizational hierarchy; Apollo mission control SPAN Trench backroom procedures calculations documentation

REAL-TIME TELEMETRY DATA FLOW — Apollo telemetry data flow MSFN Goddard Real-Time Computer Complex MOCR displays; NASA Manned Spacecraft Center Real Time Computer Complex RTCC CRT display generation 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)